

Machine Learning I: Hands-On 3 - Worked Solutions

3.1 k-nearest neighbors and Naive Bayes: worked solutions

3.1.1 Numerical k-NN

The query is

$$q = (6, 5),$$

and the Manhattan distance is

$$d_i = |\text{Age}_i - 6| + |\text{Acidity}_i - 5|.$$

Compute the distance for every observation:

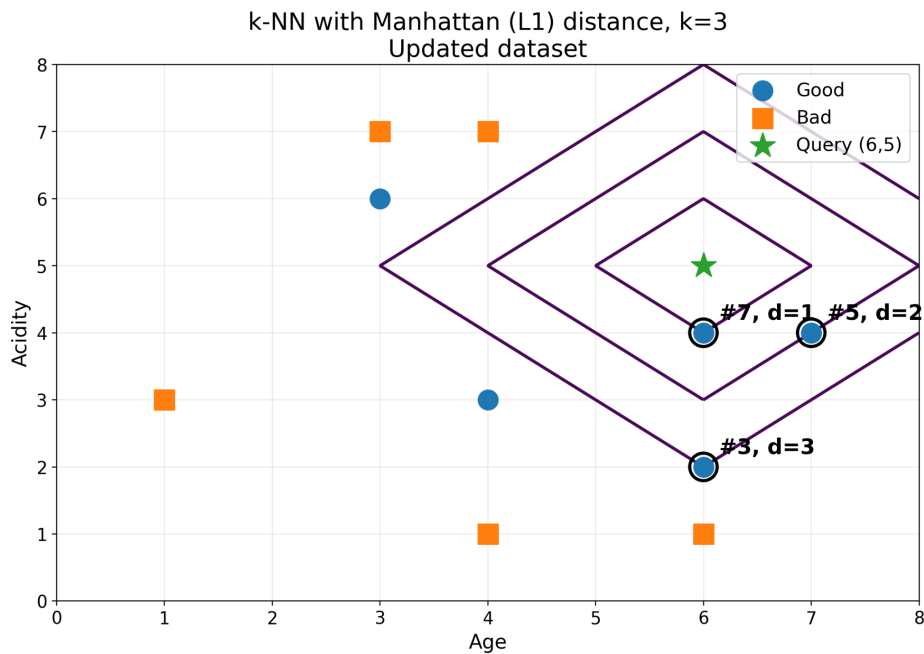
Example	Point	Quality	Calculation	Distance
1	(4, 3)	Good	$ 4 - 6 + 3 - 5 = 2 + 2$	4
2	(1, 3)	Bad	$ 1 - 6 + 3 - 5 = 5 + 2$	7
3	(6, 2)	Good	$ 6 - 6 + 2 - 5 = 0 + 3$	3
4	(3, 7)	Bad	$ 3 - 6 + 7 - 5 = 3 + 2$	5
5	(7, 4)	Good	$ 7 - 6 + 4 - 5 = 1 + 1$	2
6	(4, 1)	Bad	$ 4 - 6 + 1 - 5 = 2 + 4$	6
7	(6, 4)	Good	$ 6 - 6 + 4 - 5 = 0 + 1$	1
8	(4, 7)	Bad	$ 4 - 6 + 7 - 5 = 2 + 2$	4
9	(3, 6)	Good	$ 3 - 6 + 6 - 5 = 3 + 1$	4
10	(6, 1)	Bad	$ 6 - 6 + 1 - 5 = 0 + 4$	4

Ordering by distance gives

$$7 (1), \quad 5 (2), \quad 3 (3), \quad 1, 8, 9, 10 (4), \quad 4 (5), \quad 6 (6), \quad 2 (7).$$

For $k = 3$, the neighbors are Examples 7, 5, and 3. All three are **Good**, so

$$\hat{y} = \text{Good}.$$



The tie at distance 4 does not affect the $k = 3$ prediction, but it matters if the neighbor cutoff reaches that distance. A practical implementation must use a deterministic policy for equal distances.

As k increases, each prediction averages information over a larger neighborhood. This generally smooths the decision regions: very small k can follow local sample irregularities closely, whereas larger k tends to increase bias and reduce variance.

3.1.2 Categorical k-NN

For the query

$q = (\text{Red, SUV, Domestic, Automatic, Gasoline}),$

we count one unit of distance for every mismatching categorical feature.

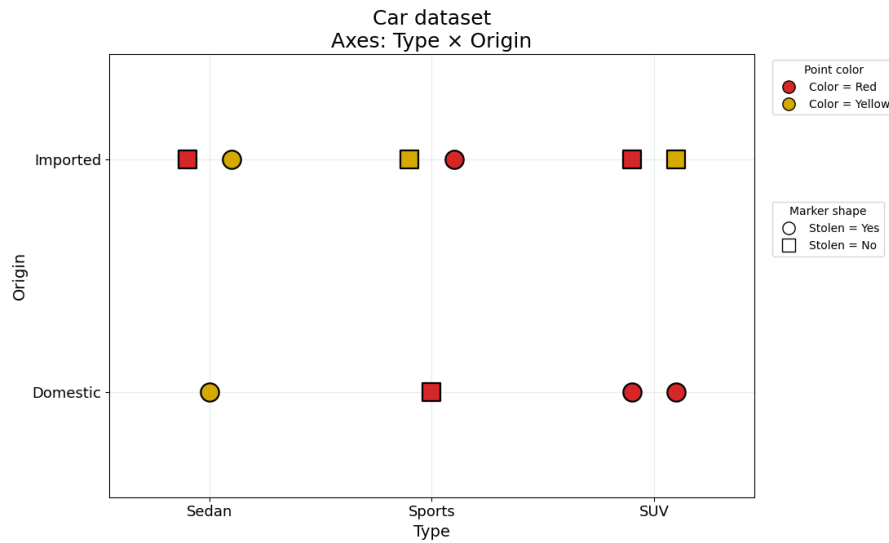
Example	Stolen?	Mismatches	Distance
1	Yes	none	0
2	Yes	Fuel	1
3	Yes	Origin, Transmission	2
4	No	Origin, Transmission	2
5	No	Type, Origin, Transmission, Fuel	4
6	Yes	Color, Type, Origin, Transmission, Fuel	5
7	Yes	Color, Type, Origin, Transmission, Fuel	5
8	No	Color, Origin, Fuel	3
9	Yes	Type, Origin, Fuel	3
10	No	Color, Type, Transmission, Fuel	4
11	Yes	Color, Type, Transmission, Fuel	4

The first two nearest neighbors are Examples 1 and 2, both **Yes**. The third-neighbor position is tied between Examples 3 and 4 at distance 2.

If ties are resolved by original row order, Example 3 is selected and the $k = 3$ vote is three **Yes** votes. If all observations tied at the cutoff are considered, the vote is three **Yes** versus one **No**. Either way, this query is predicted as

Stolen = Yes.

The 2D Type-versus-Origin projection is useful for intuition but cannot determine the actual prediction on its own because **Color, Transmission, and Fuel also contribute to the distance**. Two observations that overlap in the displayed projection may still be separated in the full five-dimensional feature space.



3.1.3 Naive Bayes

There are eight observations:

- 3 with class **Yes**;
- 5 with class **No**.

Therefore the class priors are

$$P(\text{Yes}) = \frac{3}{8}, \quad P(\text{No}) = \frac{5}{8}.$$

Step 1: Probabilities conditioned on Yes

Among the three stolen cars:

- all 3 are Red;
- all 3 are SUVs;
- 2 of the 3 are Domestic.

Hence

$$P(\text{Red} \mid \text{Yes}) = 1,$$

$$P(\text{SUV} \mid \text{Yes}) = 1,$$

$$P(\text{Domestic} \mid \text{Yes}) = \frac{2}{3}.$$

The unnormalized Naive Bayes score is

$$\begin{aligned} S_{\text{Yes}} &= P(\text{Yes})P(\text{Red} \mid \text{Yes})P(\text{SUV} \mid \text{Yes})P(\text{Domestic} \mid \text{Yes}) \\ &= \frac{3}{8} \times 1 \times 1 \times \frac{2}{3} \\ &= \frac{1}{4} = 0.25. \end{aligned}$$

Step 2: Probabilities conditioned on No

Among the five non-stolen cars:

- 2 are Red;
- 2 are SUVs;
- 2 are Domestic.

Thus

$$P(\text{Red} \mid \text{No}) = P(\text{SUV} \mid \text{No}) = P(\text{Domestic} \mid \text{No}) = \frac{2}{5}.$$

The score for class **No** is

$$\begin{aligned} S_{\text{No}} &= \frac{5}{8} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} \\ &= 0.04. \end{aligned}$$

Step 3: Compare the class scores

Since

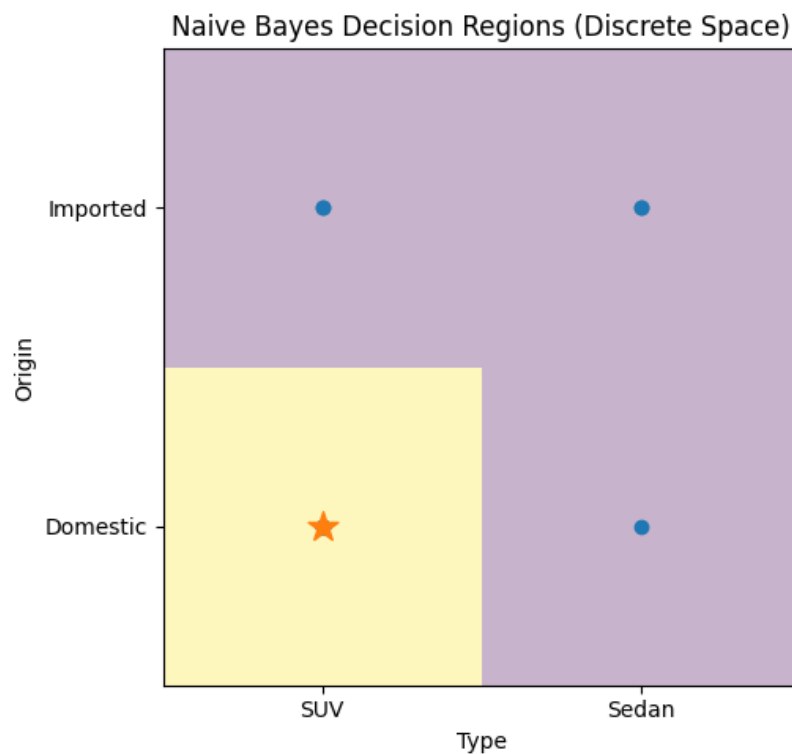
$$0.25 > 0.04,$$

Naive Bayes predicts

Stolen = Yes.

The numbers 0.25 and 0.04 are proportional to the posterior probabilities. For classification, their common normalization constant is unnecessary because only their relative magnitude determines the predicted class.

The same calculation can be repeated for every possible discrete feature combination. This produces a global partition of the discrete feature space: each combination is assigned to whichever class receives the larger posterior score.



ML insight: why k-NN and Naive Bayes can disagree

k-NN asks which labelled observations are locally closest to the query. Naive Bayes asks which class, according to a global probabilistic model, is more likely to produce the observed feature values. The algorithms therefore encode different inductive biases even when trained on exactly the same data.